

CBME Diabetes Training Module Assessment Framework

Competency-Based Assessment Rubric

Assessment Philosophy

This module follows CBME principles with **progressive entrustment**. Students progress from "shows how" to "does autonomously" based on demonstrated competence rather than time-based progression.

Entrustment Scale:

1. **I have no concern** - Student can perform independently
 2. **I have minor concern** - Student needs occasional supervision
 3. **I need to discuss with student** - Significant supervision required
 4. **I need to do it** - Direct faculty intervention needed
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Assessment Timeline & Structure

Weekly Formative Assessment

- **Monday:** Self-reflection logs (competency self-rating)
- **Wednesday:** Peer assessment of group activities
- **Friday:** Faculty mini-CEX (Clinical Evaluation Exercise)

Mid-Module Assessment (Day 5)

- **Written:** Diabetes knowledge quiz
- **OSCE:** 3 stations (screening, education, referral)
- **Portfolio:** Case logs and reflection journal

End-of-Module Assessment (Day 10)

- **Community Project:** Group presentation and evaluation
 - **Integrated OSCE:** 5 stations covering all competencies
 - **Portfolio Defense:** Individual competency demonstration
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Detailed Assessment Rubric

DOMAIN 1: PATIENT CARE (PC) - 30% Weight

PC1: Community Screening & Diagnosis

Behaviors Assessed:

- Consent obtaining for screening procedures
- Correct glucometer technique and interpretation
- Appropriate counseling based on results
- Identification and management of complications

Entrustment Scale:

- **Level 4:** Requires direct supervision for all steps
- **Level 3:** Shows correct technique with supervision
- **Level 2:** Does independently with occasional prompting
- **Level 1:** Performs autonomously in community setting

Assessment Methods: Direct observation during screening camps

PC2: Patient Education & Counseling**Behaviors Assessed:**

- Use of teach-back method for education verification
- Cultural adaptation of educational materials
- Family involvement in counseling sessions
- Adjustment of communication based on patient feedback

Criteria:

- **Communication:** Clear, empathetic, jargon-free
- **Content Accuracy:** Evidence-based information provided
- **Patient Response:** Adapts to patient's emotional state
- **Family Integration:** Appropriate inclusion of support systems

PC4: Complication Assessment & Prevention**Behaviors Assessed:**

- Systematic foot examination technique
- Recognition of diabetes-related emergencies
- Preventive care recommendations
- Quality assurance of screening services

DOMAIN 2: MEDICAL KNOWLEDGE (MK) - 15% Weight**MK1-MK4: Diabetes Pathophysiology & Management**

Assessment Format: Multiple-choice questions + short answer cases

Content Areas:

- Pathophysiology of type 1/type 2/gestational diabetes
- Diagnostic criteria and screening recommendations
- Management guidelines (ADA/IDF/Indian protocols)

- Complications spectrum and prevention strategies

Performance Scales:

- **80-100%:** Comprehensive understanding with clinical application
 - **70-79%:** Good grasp with minor gaps in application
 - **60-69%:** Basic knowledge sufficient for daily practice
 - **<60%:** Significant knowledge gaps requiring remediation
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DOMAIN 3: PRACTICE-BASED LEARNING (PBLI) - 10% Weight**PBLI1-PBLI2: Evidence-Based Practice & Self-Learning****Assessment Methods:**

- **Literature Search:** "Find evidence for metformin in pre-diabetes"
- **Critical Appraisal:** Case-based evidence evaluation
- **Self-Directed Learning Portfolio:** Learning objectives, progress tracking
- **Reflection Journals:** Daily reflections on patient encounters

Competency Indicators:

- Identifies knowledge gaps during patient care
 - Uses appropriate resources for information gathering
 - Applies evidence to clinical decision-making
 - Maintains learning portfolio with reflections
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DOMAIN 4: COMMUNICATION SKILLS (CS) - 15% Weight**CS1-CS3: Patient-Doctor-Community Communication****Assessment in Multiple Settings:**

- **Role-Play Scenarios:** SPIKES protocol for breaking bad news
- **Patient Encounters:** Real-time observation in screening camps
- **Group Presentations:** Community project presentations
- **Family Counseling:** Multi-person communication dynamics

SMART Communication Criteria:

- **S**etting: Appropriate private environment
 - **M**essage: Clear, concise communication
 - **A**ctive listening: Non-verbal cues, paraphrasing
 - **R**esponse: Addresses patient concerns
 - **T**iming: Appropriate pacing, time management
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DOMAIN 5: PROFESSIONALISM (PRO) - 10% Weight

PRO1-PRO2: Ethical Practice & Cultural Competence

Assessment Areas:

- **Confidentiality:** Patient data protection during community activities
- **Informed Consent:** Clear explanation and voluntary participation
- **Cultural Sensitivity:** Language adaptation, belief system respect
- **Professional Demeanor:** Appropriate attire, punctuality, respect

Behavioral Indicators:

- Maintains patient privacy in rural settings
 - Adapts communication to local dialects and literacy levels
 - Shows respect for traditional healing practices
 - Demonstrates empathy across socioeconomic divides
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DOMAIN 6: SYSTEMS-BASED PRACTICE (SBP) - 15% Weight

SBP1-SBP3: Healthcare System Navigation

Assessment Methods:

- **PHC Visit Simulation:** Demonstration of referral process
- **Resource Mapping:** Identification of community support systems
- **Quality Improvement:** Data analysis from screening camps
- **Advocacy Skills:** Presentation of community health needs

Competency Framework:

- Understands village health system structure
 - Identifies appropriate referral pathways
 - Advocates for patient's social determinants of health
 - Participates in community health improvement initiatives
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OSCE Station Blueprint

Station 1: Community Screening (PC1)

Task: Perform screening on standardized patient **Time:** 10 minutes **Marks:** 20 points **Critical Actions:**

1. Obtain informed consent ✓
2. Perform glucometer correctly ✓
3. Interpret results accurately ✓
4. Provide appropriate counseling ✓

Station 2: Patient Education (PC2)

Task: Educate patient about lifestyle modification **Time:** 12 minutes **Marks:** 25 points **Competencies:**

- Communication skills ✓
- Content accuracy ✓
- Cultural adaptation ✓
- Family involvement ✓

Station 3: Referral Management (SBP1)

Task: Manage patient requiring specialist care **Time:** 8 minutes **Marks:** 15 points **Checklist:**

- Identify appropriate level of care ✓
- Complete referral documentation ✓
- Explain process to patient ✓
- Provide immediate management advice ✓

Station 4: Emergency Recognition (PC4)

Task: Assess patient with possible DKA **Time:** 10 minutes **Marks:** 20 points **Critical Skills:**

- History taking for emergencies ✓
- Physical examination for complications ✓
- Emergency management initiation ✓
- Referral coordination ✓

Station 5: Community Project (SBP3)

Task: Present community intervention plan **Time:** 15 minutes **Marks:** 25 points **Assessment Areas:**

- Project planning and feasibility ✓
- Cultural appropriateness ✓
- Evaluation methodology ✓
- Resource sustainability ✓

Portfolio Assessment Framework

Required Components:

1. Daily Learning Logs (30 entries)

- Patient encounter summaries
- Learning objectives identification
- Competency self-assessment
- Reflection on improvement areas

2. Case Documentation (15 complex cases)

- History, examination, diagnosis summary
- Management decisions with rationale
- Follow-up and outcomes
- Evidence-based references

3. Special Projects Portfolio

- Community screening camp report
- Role-play performance video
- Patient education material adaptation
- Quality improvement initiative

4. Self-Assessment Matrix

- Weekly competency rating with justification
- Progress tracking across domains
- Faculty feedback incorporation
- Remediation plan identification

Grading and Progression

Competency-Based Grading:

Overall Performance	Entrustment Level	Grade	Progression
≥80% in all domains	Level 1-2	Pass with Honors	Entrustable for independent practice
≥70% in all domains	Level 2-3	Pass	Entrustable with supervision
60-69% in ≥4 domains	Level 3-4	Conditional Pass	Additional training required
<60% in any domain	Level 4	Fail	Repeat module required

Remediation Strategy:

- **Focused Training:** Additional sessions for weak competencies
- **Mentor Assignment:** Individual faculty supervision
- **Portfolio Enhancement:** Additional documentation requirements
- **Reassessment:** Target competency evaluation within 2 weeks

Quality Assurance Measures

Standard Setting:

- **Modified Angoff Method:** Expert judgment for passing standards
- **Borderline Regression:** Statistical method for cut-score determination

Reliability Enhancement:

- **Multiple Assessors:** 2-3 faculty per OSCE station
- **Calibration Exercises:** Monthly faculty training sessions
- **Inter-rater Reliability:** Checked every 3 months

Validity Evidence:

- **Content Validity:** Curriculum mapping to competencies
 - **Response Process:** Think-aloud protocols during assessments
 - **Internal Structure:** Cronbach's alpha analysis
 - **Relationship to Variables:** Correlation with practice performance
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Assessment Documentation

Digital Platform Requirements:

- **E-portfolio System:** Continuous documentation
- **OSCE Software:** Automated scheduling and scoring
- **Dashboard Analytics:** Progress monitoring
- **Feedback Integration:** Real-time performance insights

Record Keeping:

- **Individual Assessment Logs:** Detailed performance records
 - **Aggregate Data Analysis:** Program-level quality improvement
 - **Stakeholder Feedback:** Student, faculty, and community input
 - **Longitudinal Tracking:** Competency development over time
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